

Unlocking Engagement: A Data-Driven Approach for Facebook Live Sellers

Project Title: Data Analysis of Facebook Live Sellers in Thailand

Dataset: Facebook Live Sellers in Thailand Dataset

Role: Data Analyst / ML Engineer



Problem Statement: Decoding Facebook Engagement

Why Facebook Engagement Matters: In the bustling landscape of e-commerce, especially for live sellers, Facebook engagement directly translates to visibility, customer interaction, and ultimately, sales. Understanding what drives this engagement is paramount for success.

What We Aim to Analyze and Predict: This project aims to dissect the intricate patterns within Facebook posts to identify key drivers of engagement. We seek to predict which post characteristics lead to higher interaction rates.



The Utility of Clustering and Correlation:

- Clustering helps categorize posts into distinct engagement profiles, revealing hidden structures.
- Correlation analysis quantifies relationships between different engagement metrics, guiding strategic content creation.

Source: The dataset comprises publicly available Facebook posts from various live sellers operating in Thailand.

Number of Attributes: 14 attributes after initial cleaning and feature engineering.

Type of Data: Tabular data representing social media engagement metrics.

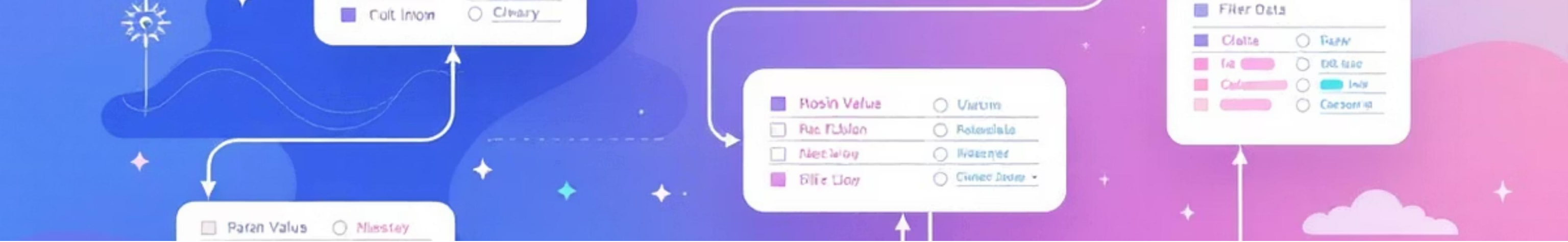


Key Attributes for Engagement Analysis

Understanding each attribute is crucial for interpreting engagement patterns. These metrics collectively paint a picture of how users interact with content.

status_type	The type of content posted (e.g., video, photo, status, link).
num_reactions	Total count of all reactions (likes, loves, wows, hahas, sads, angrys).
num_comments	Total number of comments received on the post.
num_shares	Total number of times the post was shared.
num_likes	Count of "like" reactions.
num_loves	Count of "love" reactions.
num_wows	Count of "wow" reactions.
num_hahas	Count of "haha" reactions.
num_sads	Count of "sad" reactions.
num_angrys	Count of "angry" reactions.





Data Preprocessing: Preparing for Analysis

Robust data preprocessing is essential to ensure the quality and reliability of subsequent analysis, especially for machine learning models.

Handling Missing Values

Missing values, if any, were addressed through imputation or removal to maintain dataset integrity and prevent biases in the analysis.

Removing Redundant Columns

Columns that provided duplicate information or were irrelevant to the research questions were identified and removed, streamlining the dataset for efficiency.

Normalization for K-Means

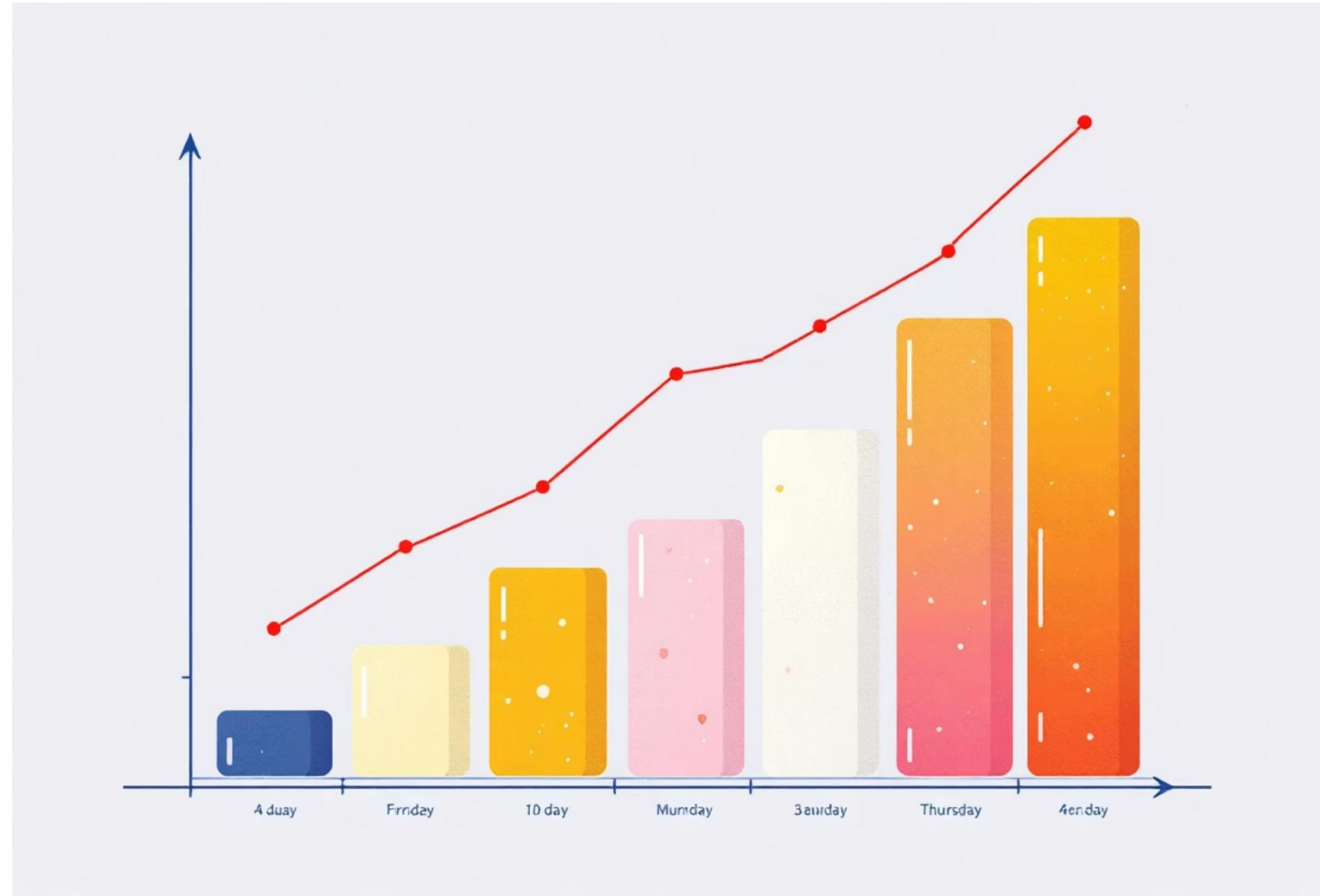
Normalization was applied to numerical features. This is critical for K-Means clustering as it relies on distance metrics, and unscaled data can lead to features with larger ranges dominating the distance calculation.

Time vs. Reactions Analysis: The Optimal Posting Day

Converting status_published: The status_published timestamp was extracted and converted into the day of the week, allowing us to analyze engagement trends based on the day.

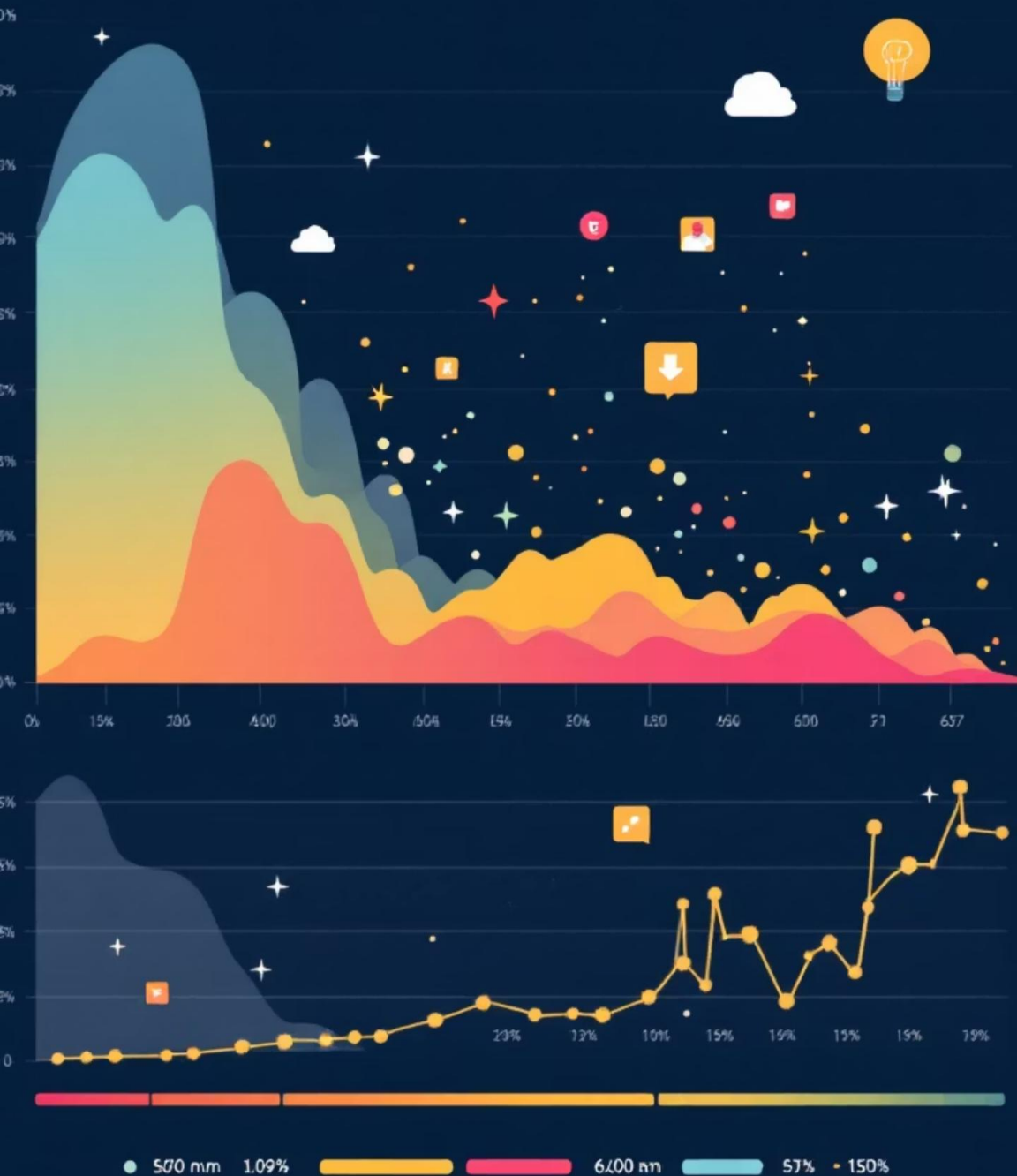
Thursday's Peak Performance: Our analysis reveals that Thursday consistently garners the maximum number of reactions, indicating a peak in audience engagement on this specific day.

Business Interpretation: Facebook live sellers should prioritize scheduling their most important content or live sessions on Thursdays to leverage the highest potential for audience interaction and visibility. This strategic timing can significantly boost campaign effectiveness.



Strong positive: content management in AM

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Correlation Analysis: Interconnected Engagement Metrics

Understanding how engagement metrics relate to each other provides insights into holistic content performance.

- **Strong Positive Correlation:** We observe a strong positive correlation between `num_reactions`, `num_comments`, and `num_shares`. This indicates that posts that receive a high number of reactions also tend to generate more comments and shares.
- **Engagement Metrics Rise Together:** This pattern suggests a symbiotic relationship: when one form of engagement increases, others tend to follow. Content that resonates strongly tends to elicit multiple forms of user interaction.
- **Implication:** Optimizing content to encourage one type of engagement (e.g., a strong call to action for comments) is likely to have a positive spillover effect on other metrics.

Clustering Goal: Segmenting Facebook Posts



Why K-Means Clustering is Used: K-Means is an unsupervised machine learning algorithm ideal for partitioning a dataset into a predefined number of clusters. It's chosen for its efficiency in grouping similar data points.

Clustering Facebook Posts: The objective is to group Facebook posts based on their engagement characteristics. This will allow us to identify distinct segments of posts that perform similarly, even if their content varies.

Business Value: By categorizing posts, live sellers can gain actionable insights. They can identify what types of posts consistently achieve high engagement, understand the characteristics of low-performing posts, and tailor future content strategies to maximize impact.

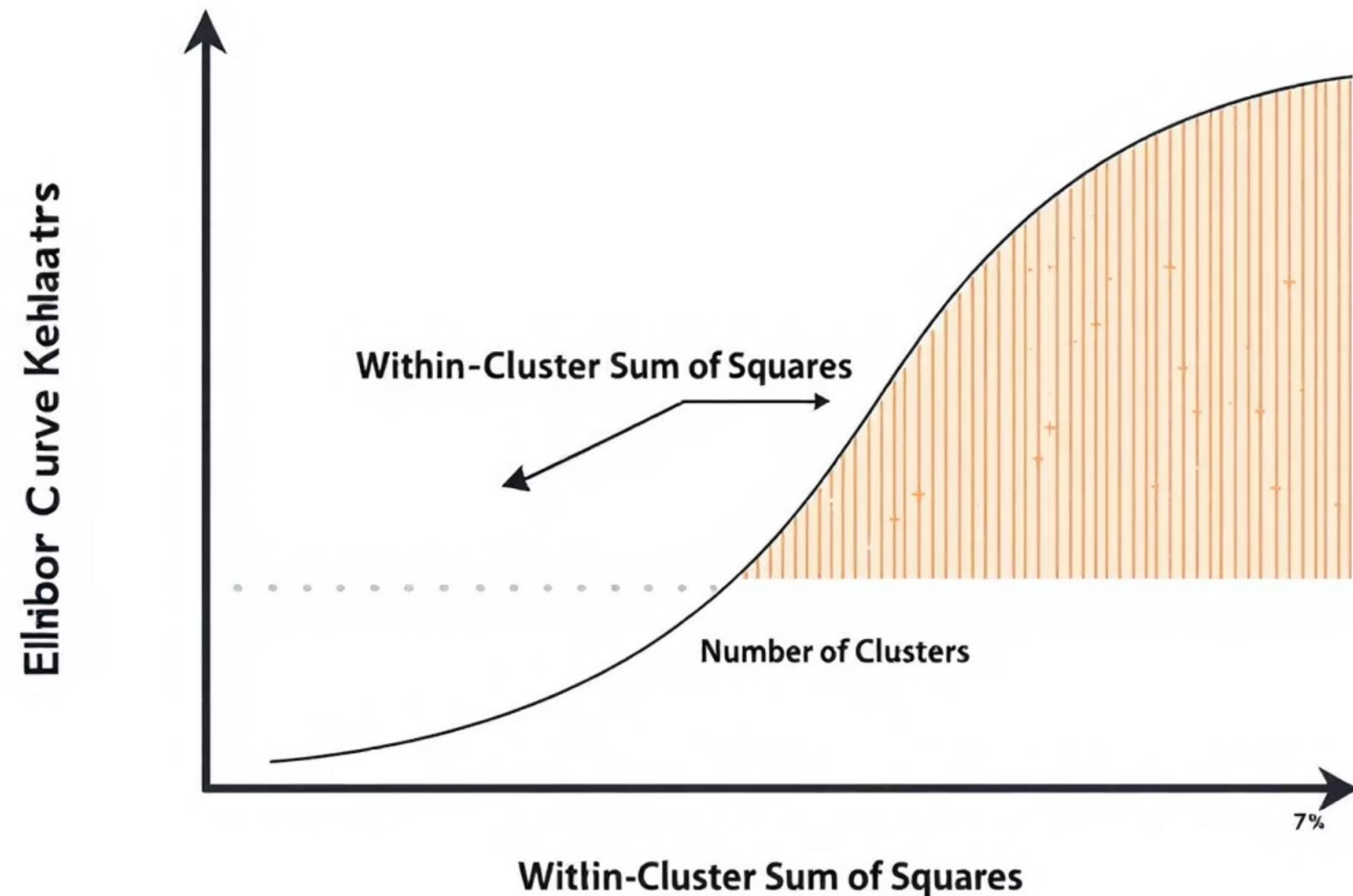


Elbow Method: Determining Optimal Clusters

Interpreting the Elbow Curve: The elbow method plots the Within-Cluster Sum of Squares (WCSS) against the number of clusters (k). As 'k' increases, WCSS generally decreases. The "elbow point" is where the rate of decrease sharply changes, resembling an arm's elbow.

Choosing Optimal 'k': This elbow point signifies the optimal number of clusters, as adding more clusters beyond this point does not significantly reduce the variance within clusters.

Avoiding Overfitting: Selecting an optimal 'k' using the elbow method prevents both underfitting (too few clusters, losing detail) and overfitting (too many clusters, capturing noise). It ensures a meaningful and robust clustering solution.



K-Means Clustering Results: Engagement Profiles

Our K-Means analysis successfully segmented the Facebook posts into distinct engagement groups, providing a clear categorization of content performance.



High-Engagement Posts

These posts are characterized by a high number of reactions, comments, and shares. They represent content that strongly resonates with the audience.



Moderate-Engagement Posts

Posts in this cluster show average levels of interaction. There's potential to optimize these to move into the high-engagement category.



Low-Engagement Posts

These posts typically receive minimal reactions. Analyzing their common attributes can help identify areas for improvement or content to avoid.

